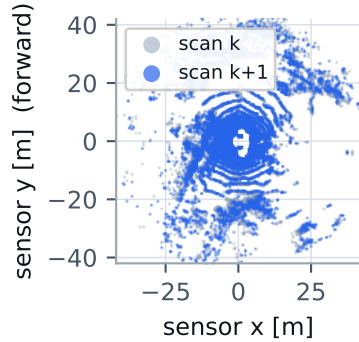


Any odometry, one interface

Four front ends of three different sensing principles are reduced to the same per-frame twist (v, ω) . Everything downstream reads only that.

Ford Multi-AV · Log 4 · 20 s window
LiDAR — scan-to-scan registration

KISS-ICP on the RED HDL-32E, deskewed · $t = 18.1$ s

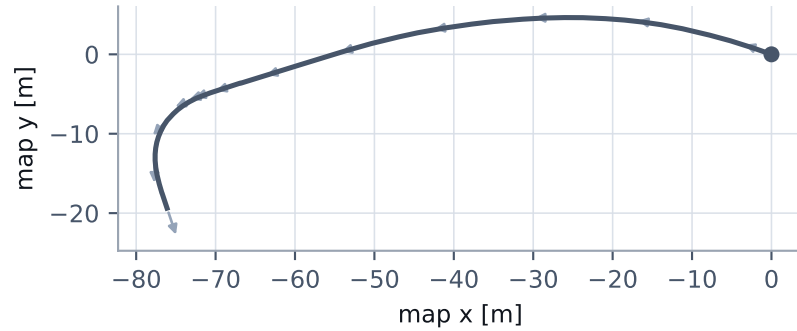


$\Delta t = 103$ ms $\rightarrow |v| = 4.62$ m/s, $\omega = -0.571$ rad/s

paper front end

GNSS/INS — pose differencing

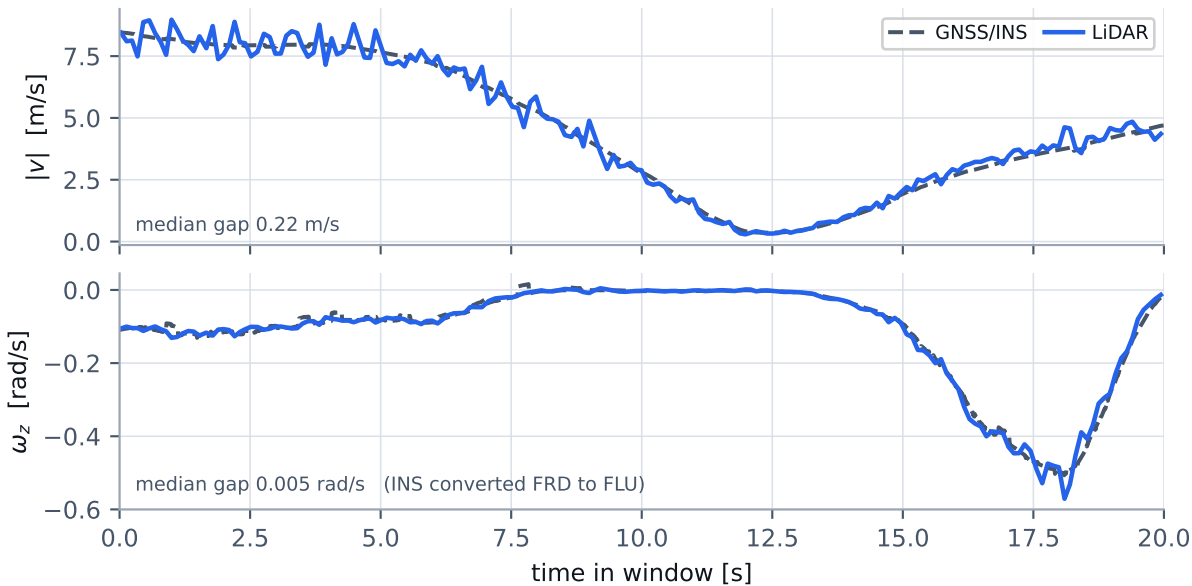
6-DoF pose stream at 200 Hz, FRD body frame



$(p_k, R_k) \rightarrow (v, \omega)$ by finite difference · heading sweep 129° over the window

same interface

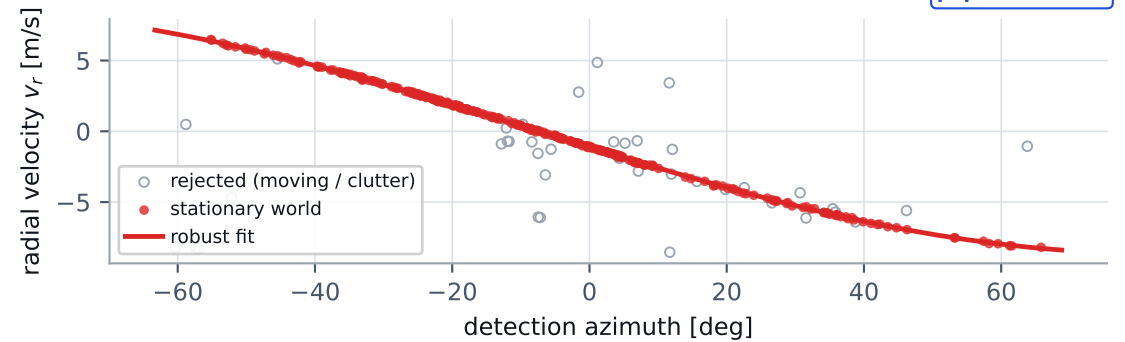
Common twist interface consumed by the calibration



Per-sensor components differ by mounting — that difference is exactly what the calibration estimates; the magnitudes above are frame-independent and therefore comparable.

RadarScenes · sequence 1 · 20 s window
Radar — Doppler ego-velocity

one 2-D radar scan, sensor 1, 324 detections

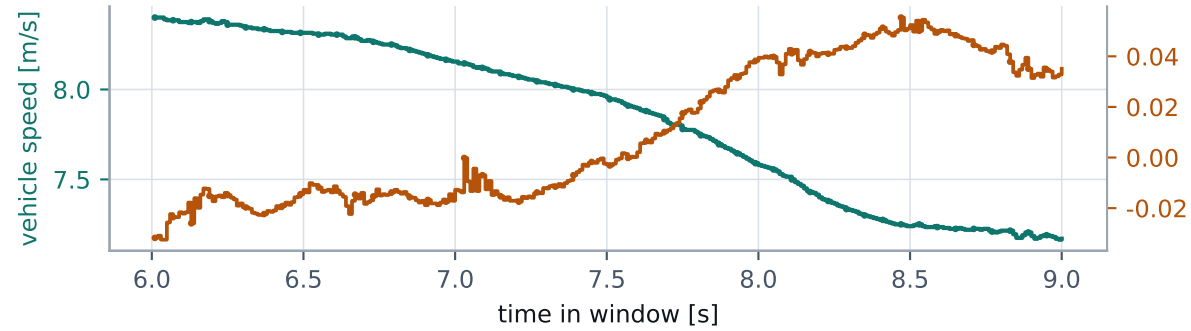


$v_r = -(v_x \cos \alpha + v_y \sin \alpha) \rightarrow |v| = 8.63$ m/s (ω is not observable from one 2-D radar)

paper front end

CAN — bus messages

two scalar signals at 100 Hz · 3 s excerpt



no point cloud, no matching — the two bus signals are already the twist

same interface

